Erisma — User manual

Version 0.4.0

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1. What Erisma is

Erisma is a suite of structural and geotechnical calculations to the Eurocodes. Every calculation is written out step by step — **input → formula → value → clause → check** — so that the engineer can verify each number by hand and submit the calculation sheet.

- **Results while you type:** checks, report and 3D model update after every change.
- **Worked examples as defaults:** every calculation opens with realistic values that pass, so you always start from a sensible design and change only what differs.
- **Hints:** every failing check says what to change (“increase the base width…”).
- **National Annexes:** partial factors and nationally determined parameters are read from the annex chosen in the project details and listed in every report.
- **Verified:** each module is tested against hand calculations and published values.

> Erisma is a tool for qualified engineers. The engineer remains responsible for the model, the input values and the interpretation of the results.

2. Installation and start

1. Install Python 3.10 or newer (on Windows tick *Add Python to PATH*).
2. Unzip `Erisma.zip` into a folder of your choice.
3. **Windows:** double-click `Erisma.bat`. **Mac:** right-click `Erisma.command` → Open.

The first start installs the program (1–2 minutes, internet needed once). Erisma then opens by itself in your browser at `http://127.0.0.1:8000`. Everything runs on your own computer and works offline; nothing is sent to the internet. To stop Erisma close the black window.

3. The screen

- **Home page:** all calculations as cards grouped by category, with search. “Continue” reopens the last calculation.
- **Left sidebar:** search (`Ctrl`+`K`), ★ favourites, recent calculations and all categories. « collapses it to icons.
- **Project details** (top of the form): company, project, job no., element, engineer, checker, date, **National Annex** and **project site**. They appear in the title block of every report and are kept for all calculations.
- **Input form** (middle): fields grouped in boxes. Above it the **status banner**: green when all checks pass, red with the failing checks and what to change.
- **Report / 3D model** (right): the full calculation sheet and the interactive model.
- Top bar: Home, light/dark theme, language ΕΛ/EN, Open, Save, Export HTML, Print / PDF, Manual.

4. Entering data

- **Decimals:** type `1,5` or `1.5` — both work.
- **Arithmetic:** type `2*3,5`, `(4+2)/2` or `1,5^2`; the result is shown next to the field and replaces the expression when you leave it.
- **Units** are shown next to every field; type the value in that unit.
- **Keyboard:** `Enter` goes to the next field, `Shift`+`Enter` back; `↑`/`↓` change the value in small steps (with `Shift` ×10).
- **Help:** the help text of a field appears while the field is selected, or click ⓘ.
- **Changed fields** are marked with a dot; ↺ returns that field to the example value. Each group shows how many of its fields you changed. “Reset to defaults” restores the whole example.
- **Groups** can be collapsed by clicking their title — Erisma remembers it.
- **Conditional fields** appear only when needed (e.g. jacket data only when “RC jacket” is chosen).
- **Tables** (soil layers, anchors, loads…): “+ Add row” copies the last row, ⧉ duplicates a row, × deletes it. You can **paste a block of cells from Excel**: select the first cell and press `Ctrl`+`V`.
- **Undo / redo:** `Ctrl`+`Z` / `Ctrl`+`Y`.
- **Wrong values** turn red immediately; if the calculation cannot run, the banner says why and the field concerned is highlighted.

5. Saving, files and printing

- **Autosave:** the inputs of every calculation are kept in the browser; when you reopen a calculation you continue where you left off.
- **Save** (`Ctrl`+`S`) downloads a project file `.erisma` with the inputs and the project details; **Open** (`Ctrl`+`O`) loads it again — send it to a colleague or keep it with the project.
- **Print / PDF** (`Ctrl`+`P`) prints the report; choose “Save as PDF” as the printer to get a PDF. **Export HTML** saves the report as a web page.

6. National Annex and project site

Choose the **National Annex** in the project details. All partial factors and nationally determined parameters (NDPs) come from it, and each report ends with a table of the NDPs used. Where Erisma has no annex for a part (for example EN 1995-1-1 for Cyprus) the EN recommended values are used and the report says so.

With the Cyprus annex, 📍 **Map** opens the official seismic zonation map: click on the site, search a town, use your location or type coordinates. The seismic zone and a_{gR} are filled in automatically (near a boundary the higher zone within 1 km is used), and the wind and snow calculation reads $v_{b,0}$ and s_k for the same site. Next to every a_{gR} field the zone buttons set the value with one click.

7. Reading the report

- The **title block** carries the project details; the summary lists every check with its utilisation.
- Each line shows **symbol = expression = value unit** with the clause of the standard.
- Checks show demand ≤ capacity, the utilisation U and **PASS / FAIL**; failing checks carry a hint in the status banner.
- Figures (sections, pressure diagrams, spectra) are drawn to scale from the actual values.
- The last table lists the National Annex parameters used.

8. The 3D model

Every calculation has an interactive 3D model (tab “3D model” or `Ctrl`+`3`). Drag to rotate, right-drag to pan, scroll to zoom. The **Layers** panel switches soil, loads, pressures, reinforcement, deflected shapes and labels on and off; the buttons give 3D, front and plan views and **PNG** saves an image for your report. The panel on the right lists the utilisation of every check; the model updates with every change.

9. Keyboard shortcuts

Keys	Action
`Ctrl+K`	Search calculations
`Ctrl+S`	Save project file
`Ctrl+O`	Open project file
`Ctrl+P`	Print / PDF
`Ctrl+3`	Report ↔ 3D model
`Ctrl+Z / Ctrl+Y`	Undo / redo inputs
`Enter / Shift+Enter`	Next / previous field
`↑ / ↓ (Shift)`	Change the value (×10)
`F1`	This manual

10. National Annexes in Erisma

EN recommended values

EN 1990/1992/1997/1998 recommended values

Cyprus (CYS National Annexes)

NA to CYS EN 1990:2002, EN 1992-1-1:2004, EN 1997-1:2004, EN 1998-1:2004, EN 1998-5:2004 (approved 11/06/2010)

- Design Approach 2 is mandatory for STR/GEO (NA 2.9).
- Seismic zones: Zone 1 0.15 g, Zone 2 0.20 g, Zone 3 0.25 g (NA zonation map, 10 % in 50 years).

11. Calculation reference

ANALYSIS

Continuous beam analysis

Standard: Stiffness method

Linear elastic analysis of single or multi-span beams with any number of supports, point loads, UDLs and moments. Gives reactions, shear, moment and deflection diagrams.

Method and assumptions

- Linear elastic stiffness (finite element) method, Euler–Bernoulli beam.
- Supports pinned or fixed at any position; loads downward positive.

Inputs

Input	Unit	Example	Range	Notes
Total length	m	10	0.1 – …	
Material		Steel (E = 210 GPa)	Steel (E = 210 GPa) / Concrete C25/30 (E = 31 GPa) / Timber C24 (E = 11 GPa) / Aluminium (E = 70 GPa)	
Second moment of area I	cm^4	8356	0.001 – …	
Supports	table			Position [m], Type
Distributed loads	table			Start [m], End [m], w (down +) [kN/m]
Point loads	table			Position [m], P (down +) [kN]
Applied moments	table			Position [m], M (clockwise +) [kNm]

CONCRETE

RC beam section design

Standard: EN 1992-1-1:2004

Rectangular section with tension and compression reinforcement: flexure (stress block, strain compatibility), minimum/maximum reinforcement and shear with variable strut inclination.

Method and assumptions

- Rectangular stress block (λ = 0.8, η = 1.0), ε_cu3 = 3.5 ‰, x/d ≤ 0.45 for ductility.
- Shear with variable strut inclination 1 ≤ cot θ ≤ 2.5 (6.2.3).

Inputs

Input	Unit	Example	Range	Notes
Width b	mm	300	100 – …	
Overall depth h	mm	500	150 – …	
Nominal cover to links	mm	30	10 – …	
Concrete class		C25/30	7 …	
Reinforcement fyk	N/mm²	500	200 – 600	
γc		1.5	1 – …	
γs		1.15	1 – …	
αcc		1	0.8 – 1	1.0 recommended (Greek NA); 0.85 UK NA.
Design moment MEd	kNm	140	0 – …	
Design shear VEd	kN	120	0 – …	
Number of tension bars		4	2 – …	
Tension bar diameter		16	8 …	
Number of compression (top) bars		2	2 – …	
Compression bar diameter		12	8 …	
Link diameter		8	8 / 10 / 12 / 14	
Link legs		2	2 – …	
Link spacing	mm	200	50 – …	

Checks performed (worked example)

Check	Clause	U (example)
Tension reinforcement: A_s,prov ≥ A_s,req	6.1	0.95
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1(1)	0.23
Maximum reinforcement: A_s + A_s2 ≤ A_s,max	9.2.1.1(3)	0.17
Bar spacing: s_clear ≥ max(φ, d_g + 5, 20)	8.2(2)	0.47
Moment resistance: M_Ed ≤ M_Rd	6.1	0.96
Concrete strut: V_Ed ≤ V_Rd,max	6.2.3(3)	0.32

Shear reinforcement: V_Ed ≤ V_Rd,s	6.2.3(3)	0.54
Minimum links: A_sw/(s b) ≥ ρ_w,min	9.2.2(5)	0.48
Link spacing: s ≤ 0.75d	9.2.2(6)	0.59

RC column

Standard: EN 1992-1-1 §5.8, EN 1998-1 §5.4

Rectangular column: imperfections, slenderness, second-order effects (nominal curvature), biaxial bending (5.8.9), shear, detailing and EC8 DCM rules. Interactive 3D with reinforcement.

Method and assumptions

- Second-order effects by the nominal curvature method (5.8.8); biaxial by Eq. 5.39.
- M_Rd by strain compatibility for the actual axial force; EC8 DCM rules optional.

Inputs

Input	Unit	Example	Range	Notes
Width b	mm	400	200 – …	
Depth h (bending about the strong axis)	mm	400	200 – …	
Cover to links	mm	30	15 – …	
Concrete class		C25/30	7 …	
Reinforcement fyk	N/mm²	500	400 – 600	
Longitudinal bars Ø		16	9 …	
Bars along b (each face, incl. corners)		4	2 – …	
Bars along h (each face, incl. corners)		4	2 – …	
Links Ø		8	8 / 10 / 12	
Link spacing	mm	100	50 – …	
Clear height	m	3	0.5 – …	
Effective length factor (bending in h)		1	0.5 – 2.5	0.5 fixed–fixed, 0.7 fixed–pinned, 1.0 pinned–pinned, 2.0 cantilever (or EN 1992-1-1 Eq. 5.15/5.16).
Effective length factor (bending in b)		1	0.5 – 2.5	
Braced column		yes		Unbraced columns use r_m = 1.0 (C = 0.7).

Axial force N_Ed (compression +)	kN	1200	0 – …	
M_Ed (h) at top	kNm	60		First-order end moments; same sign = single curvature.
M_Ed (h) at bottom	kNm	-30		
M_Ed (b) at top	kNm	20		
M_Ed (b) at bottom	kNm	-10		
Shear V_Ed	kN	60	0 – …	
Effective creep ratio φ_ef		0.8	0 – 3	
Primary seismic column (EN 1998-1 DCM)		yes		

Checks performed (worked example)

Check	Clause	U (example)
Axial: N_Ed ≤ N_Rd	6.1	0.32
Bending in h: M_Ed ≤ M_Rd	6.1	0.28
Bending in b: M_Ed ≤ M_Rd	6.1	0.12
Biaxial: (M_Ed,h/M_Rd,h)^a + (M_Ed,b/M_Rd,b)^a ≤ 1	EN 1992-1-1 Eq. 5.39	0.31
Shear: V_Ed ≤ max(V_Rd,c, min(V_Rd,s, V_Rd,max))	6.2	0.17
A_s ≥ A_s,min = max(0.10 N_Ed/f_yd, 0.002 A_c)	9.5.2(2)	0.13
A_s ≤ 0.04 A_c	9.5.2(3)	0.38
Link spacing ≤ min(20φ, b_min, 400)	9.5.3(3)	0.31
Axial load ratio v_d ≤ 0.65	EN 1998-1 5.4.3.2.1(3)	0.69
ρ_l ≥ 1 %	5.4.3.2.2(1)	0.66
ρ_l ≤ 4 %	5.4.3.2.2(1)	0.38
Critical region links: s ≤ min(b_0/2, 175, 8d_bL)	5.4.3.2.2(11)	0.78

RC slab

Standard: EN 1992-1-1

One-way continuous slabs with pattern loading, or two-way rectangular panels with any edge condition (plate analysis); bottom/top reinforcement, shear, deflection (span/depth) and punching at a column.

Method and assumptions

- Two-way panels: Kirchhoff plate by finite differences with any edge conditions (verified against Timoshenko's tables).

- One-way continuous slabs: pattern loading (all / odd / even spans).

Inputs

Input	Unit	Example	Range	Notes
Slab type		Two-way panel	Two-way panel / One-way continuous	
Short span l_x	m	4.5	1 – …	
Long span l_y (two-way)	m	6	1 – …	
Number of equal spans (one-way)		3	1 – 8	
Edge x = 0		Fixed / continuous	Simply supported / Fixed / continuous	
Edge x = l_x		Simply supported	Simply supported / Fixed / continuous	
Edge y = 0		Fixed / continuous	Simply supported / Fixed / continuous	
Edge y = l_y		Simply supported	Simply supported / Fixed / continuous	
Slab thickness h	mm	180	100 – …	
Cover	mm	25	15 – …	
Concrete class		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Superimposed permanent load (excl. self weight)	kN/m²	2	0 – …	Finishes, screed, partitions.
Imposed load	kN/m²	2	0 – …	EN 1991-1-1: residential 1.5–2.0, offices 3.0, balconies 4.0.
Bottom bars Ø		10	8 / 10 / 12 / 14 / 16	
Bottom bar spacing	mm	150	75 – 400	
Top bars Ø (supports)		12	8 / 10 / 12 / 14 / 16	
Top bar spacing	mm	150	75 – 400	
Check punching at a column (flat slab)		no		
Column size c1	mm	400	150 – …	
Column size c2	mm	400	150 – …	
Column reaction V_Ed	kN	450	0 – …	
Column position		Internal	Internal / Edge / Corner	β = 1.15 / 1.4 / 1.5 (EN 1992-1-1 Fig. 6.21N).

Checks performed (worked example)

Check	Clause	U (example)
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.35
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.38
Bending: M_Ed ≤ M_Rd	6.1	0.34
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.23
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.36
Bending: M_Ed ≤ M_Rd	6.1	0.23
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.49
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.26
Bending: M_Ed ≤ M_Rd	6.1	0.5
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.4
Minimum reinforcement: A_s,prov ≥ A_s,min	9.3.1.1	0.26
Bending: M_Ed ≤ M_Rd	6.1	0.41
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.41
Span/depth: l/d ≤ (l/d)_lim	7.4.2	0.59

Water tank / pool

Standard: EN 1992-3, EN 1992-1-1, EN 1997-1

Rectangular reinforced concrete tank or swimming pool: wall bending for the full (water test) and empty (backfill + groundwater) cases, crack width to EN 1992-3 tightness class 1, minimum thickness and reinforcement, and flotation (UPL) of the empty tank. Interactive 3D.

Method and assumptions

- Long walls as vertical strips fixed into the base, free or pinned at the top.
- Crack width by EN 1992-1-1 7.3.4 (quasi-permanent) against w_k1 of tightness class 1.
- Thermal/early-age restraint cracking is not included.

Inputs

Input	Unit	Example	Range	Notes
Internal length	m	10	1 – …	

Internal width	m	5	1 – …	
Wall height (above base)	m	2.5	0.5 – 12	
Maximum water depth	m	2.3	0.1 – …	
Wall thickness	mm	250	120 – 1000	
Base slab thickness	mm	300	150 – 1500	
Base projection beyond the walls	m	0.3	0 – …	Soil resting on the projection helps against flotation.
Wall support at the top		Free (open tank / pool)	Free (open tank / pool) / Pinned (roof slab)	
Backfill height outside (above base)	m	2.5	0 – …	
Backfill γ	kN/m³	19	10 – …	
Backfill φ'k	°	30	20 – 45	
Surcharge on the backfill	kN/m²	5	0 – …	
Groundwater outside (above base underside)	m	1	0 – …	Highest expected level — governs flotation of the empty tank.
Concrete class		C30/37	C30/37 / C35/45 / C40/50	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover to bars	mm	40	25 – 75	
Vertical bars, inside face Ø		12	10 / 12 / 14 / 16 / 20	
Spacing, inside face	mm	125	75 – 300	
Vertical bars, outside face Ø		12	10 / 12 / 14 / 16 / 20	
Spacing, outside face	mm	125	75 – 300	

Checks performed (worked example)

Check	Clause	U (example)
Minimum wall thickness (EN 1992-3 9.11.1, classes 1–2)	EN 1992-3 9.11.1(102)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.31
Bending: M_Ed ≤ M_Rd	6.1	0.31
A_s ≥ A_s,min = k_c k f_ct,eff A_ct / f_yk (k_c = 0.4)	EN 1992-1-1 7.3.2	0.32
Crack width inside: w_k ≤ w_k1	EN 1992-3 7.3.1	0.56

Compression zone x ≥ x_min = min(50 mm, 0.2t) (no through-cracks)	EN 1992-3 7.3.1(112)	0.81
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.61
Bending: M_Ed ≤ M_Rd	6.1	0.61
A_s ≥ A_s,min = k_c k f_ct,eff A_ct / f_yk (k_c = 0.4)	EN 1992-1-1 7.3.2	0.32
Crack width outside: w_k ≤ w_k1	EN 1992-3 7.3.1	0.78
Compression zone x ≥ x_min = min(50 mm, 0.2t) (no through-cracks)	EN 1992-3 7.3.1(112)	0.81
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.46
UPL: γ_G,dst U ≤ γ_G,stb G (1 / 0.9)	EN 1997-1 Eq. 2.8	0.5

EXISTING BUILDINGS

Assessment and strengthening of RC members

Standard: EN 1998-3 Annex A

Existing RC column or beam: knowledge level and confidence factor, chord rotation at yield and ultimate (A.10a, A.1), cyclic shear resistance (A.12), checks for the limit states NC, SD and DL, and strengthening with an RC jacket or FRP wrapping. Interactive 3D model.

Method and assumptions

- EN 1998-3 Annex A: θ_y (A.10a), θ_um (A.1), V_R (A.12); mean strengths ÷ CF, brittle shear also ÷ γ (Cyprus NA 2.3).
- NC: θ_E ≤ θ_um, SD: θ_E ≤ ¾θ_um, DL: θ_E ≤ θ_y. RC jacket by the monolithic factors of A.4.2.2, FRP by A.4.4. The Greek KANEPE is not implemented.

Inputs

Input	Unit	Example	Range	Notes
Knowledge level		KL2 — normal knowledge	KL1 — limited knowledge / KL2 — normal knowledge / KL3 — full knowledge	KL1: simulated design + limited in-situ tests; KL2: extended inspection and testing; KL3: comprehensive (Table 3.1).
Primary seismic element		yes		γ_el = 1.5 (deformation) and 1.15 (shear) for primary elements, 1.0 for secondary.
Detailed for earthquake resistance		no		Off for older buildings (no seismic detailing): θ_um is divided by 1.2 (A.3.2.2).
Ribbed (deformed) bars		yes		
Width b	mm	300	150 – …	

Depth h (in the direction of bending)	mm	400	150 – …	
Cover to bar centre	mm	35	15 – …	
Bars along b (each face, incl. corners)		3	2 – …	
Bars along h (each face, incl. corners)		3	2 – …	
Longitudinal bar Ø	mm	16	8 – 32	
Hoop Ø	mm	8	6 – 14	
Hoop spacing	mm	200	50 – 400	
Hoop legs parallel to the shear		2	2 – 6	
Concrete mean strength f_cm (tests)	N/mm²	16	8 – 60	
Steel mean yield strength f_ym	N/mm²	450	220 – 700	
Shear span L_V = M/V	m	1.5	0.3 – …	Half the clear height for a column in double curvature.
Axial force in the seismic situation	kN	600	0 – …	Compression positive (G + ψ2Q).
Chord rotation demand θ_E — NC		0.02	0 – 0.2	rad, from the analysis for the NC seismic action.
Chord rotation demand θ_E — SD		0.012	0 – 0.2	
Chord rotation demand θ_E — DL		0.004	0 – 0.2	
Shear demand V_E	kN	150	0 – …	From capacity design: ΣM_y / L_cl of the member, or from the analysis.
Strengthening		RC jacket	None (assessment only) / RC jacket / FRP wrapping	

Jacket thickness	mm	100	60 – 250	≥ 75 mm cast, ≥ 50 mm shotcrete.
Jacket concrete mean strength	N/mm²	33	20 – 60	
Jacket bars along each face (incl. corners)		3	2 – …	
Jacket bar Ø	mm	16	10 – 32	
Jacket hoop Ø	mm	10	8 – 14	
Jacket hoop spacing	mm	100	50 – 200	
Jacket steel mean yield strength	N/mm²	550	400 – 700	
FRP layers		2	1 – 10	
Thickness per layer	mm	0.17	0.05 – 2	
FRP modulus E_f	GPa	230	20 – 650	
FRP ultimate strain ε_u,f		0.015	0.003 – 0.05	
Corner radius	mm	25	10 – 60	Round the corners before wrapping (≥ 20–25 mm).

Checks performed (worked example)

Check	Clause	U (example)
NC (with RC jacket): θ_E ≤ θ_um	EN 1998-3 A.3.2.2	0.72
SD (with RC jacket): θ_E ≤ ¾ θ_um	EN 1998-3 A.3.2.3	0.58
DL (with RC jacket): θ_E ≤ θ_y	EN 1998-3 A.3.2.4	0.62
Shear (with RC jacket): V_E ≤ V_R	EN 1998-3 A.3.3.1	0.44

GEOTECHNICAL

Berlin wall (soldier pile wall)

Standard: EN 1997-1, EN 1993-1-1, EN 1995-1-1, EN 1998-5

Steel soldier piles with timber lagging, cantilever or with one anchor row: active/passive earth pressure with spatial passive resistance, embedment (Blum / free earth support), pile bending and

shear (EC3), lagging (EC5), anchor force, seismic case and indicative deflection. Interactive 3D model.

Method and assumptions

- Blum (cantilever) or free earth support (one anchor row); spatial passive resistance below the excavation (f·b ≤ a).

Inputs

Input	Unit	Example	Range	Notes
Excavation depth H	m	5	1 – 15	
Pile spacing a (centre to centre)	m	2	0.5 – 4	Typical 1.5–3.0 m.
Provided embedment below excavation t	m	3.5	0.5 – …	
Support system		Single anchor row	Cantilever / Single anchor row	
Anchor depth below top	m	1.5	0.3 – …	
Anchor inclination below horizontal	°	20	0 – 45	
Anchor design resistance R_a,d (per anchor)	kN	450	1 – …	From anchor design / suitability tests (EN 1997-1 §8).
Number of piles shown in 3D		6	2 – 12	
Pile section		HEB 240	40 …	
Steel grade		S355	S235 / S275 / S355	
Lagging board thickness	mm	80	30 – 200	
Lagging pressure factor (arching)		0.6	0.3 – 1	Arching reduces the pressure on the boards: ~0.6 in competent soils, 1.0 in soft clay or loose sand.
Soil φ'k	°	32	20 – 45	
Soil c'k	kN/m²	0	0 – …	
Soil γ	kN/m³	19	10 – …	
Active wall friction δa/φ'		0.667	0 – 0.667	
Passive wall friction δp/φ'		0.333	0 – 0.5	Keep ≤ 0.5: Coulomb overestimates passive resistance.

Surcharge behind the wall (variable)	kN/m²	10	0 – …	
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	
Displacement factor r		1.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: use 1.0 for anchored / propped walls.
ψ2 for surcharge		0.3	0 – 1	

Checks performed (worked example)

Check	Clause	U (example)
DA2: t_prov ≥ t_req	EN 1997-1 9.7.4	0.62
Seismic: t_prov ≥ t_req	EN 1997-1 9.7.4	0.8
Pile bending: M_Ed ≤ M_c,Rd	EN 1993-1-1 6.2.5	0.42
Pile shear: V_Ed ≤ V_pl,Rd	EN 1993-1-1 6.2.6	0.19
Embedment: t_prov ≥ t_req	EN 1997-1 9.7.4	0.8
Anchor: A_d ≤ R_a,d	EN 1997-1 8.5	0.39
Lagging: t_prov ≥ t_req	EN 1995-1-1 6.1	0.69

Cantilever retaining wall

Standard: EN 1997-1, EN 1998-5, EN 1992-1-1

Reinforced concrete cantilever wall: active earth pressure (Coulomb), surcharge, groundwater, seismic thrust (Mononobe-Okabe), sliding, overturning, bearing (Annex D), and EC2 design of stem, heel and toe. Includes an interactive 3D model.

Method and assumptions

- Coulomb active pressure on the virtual back through the heel; Mononobe-Okabe in the seismic case (±k_v); bearing by Annex D with inclination factors.
- Passive resistance in front is ignored except at a shear key.

Inputs

Input	Unit	Example	Range	Notes

Stem height above base	m	4	0.5 – 15	
Stem thickness at top	m	0.25	0.15 – …	
Stem thickness at base	m	0.4	0.15 – …	
Base thickness	m	0.5	0.2 – …	
Toe length	m	1	0 – …	
Heel length (behind stem)	m	2.2	0.2 – …	
Soil depth over toe	m	0.5	0 – …	Depth of soil in front of the wall above the top of the base.
Wall length (3D view)	m	8	1 – …	
Stepped stem (thicker lower part)		no		Thicken the lower part of the stem where the moments are largest.
Height of the step above the base	m	1.5	0.2 – …	
Stem thickness below the step	m	0.55	0.2 – …	Above the step the stem tapers from 'thickness at base' to 'thickness at top'.
Shear key under the base		no		A downstand under the base that mobilises passive resistance against sliding.
Key position (front face, from the toe)	m	1	0 – …	Usually under the stem or just behind it.
Key width	m	0.4	0.2 – …	
Key depth below the base	m	0.5	0.2 – 2	
Backfill slope β	°	0	0 – 40	
Backfill φ'k	°	30	15 – 45	
Backfill γ	kN/m³	18	10 – …	
Backfill γsat	kN/m³	20	10 – …	
Surcharge on backfill (variable)	kN/m²	10	0 – …	e.g. 10 kN/m² for light traffic or storage.
Water level above base underside	m	0	0 – …	0 = drained backfill (no water pressure).
Foundation φ'k	°	32	10 – 45	
Foundation c'k	kN/m²	0	0 – …	
Foundation γ	kN/m³	19	10 – …	
Base friction factor k (tan δ = k tan φ')		1	0.3 – 1	1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)).

Design approach		National Annex	National Annex / DA1 / DA2 / DA3	'National Annex' uses the approach of the annex chosen in the project details.
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4.
Displacement factor r		2.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped).
ψ2 for surcharge		0.3	0 – 1	
Concrete class		C30/37	C25/30 / C30/37 / C35/45 / C40/50	
Reinforcement fyk	N/mm²	500	400 – 600	
Concrete unit weight	kN/m³	25	20 – …	
Cover — stem (earth face)	mm	50	20 – …	
Cover — base	mm	75	25 – …	
Stem bars Ø		16	7 …	
Stem bar spacing	mm	150	75 – 400	
Heel top bars Ø		16	7 …	
Heel bar spacing	mm	150	75 – 400	
Toe bottom bars Ø		12	7 …	
Toe bar spacing	mm	150	75 – 400	
Shear key bars Ø		12	7 …	
Shear key bar spacing	mm	200	75 – 400	

Checks performed (worked example)

Check	Clause	U (example)
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.73
DA2 eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.33
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.42
Overturning: M_dst,d ≤ M_stb,d	EN 1997-1 Eq. 2.4	0.39
Seismic sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.92

Seismic eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.4
Seismic bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.8
Seismic overturning: M_dst ≤ M_stb	EN 1998-5 7.3.1	0.37
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.67
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.38
Bending: M_Ed ≤ M_Rd	6.1	0.66
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.51
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.5
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.47
Bending: M_Ed ≤ M_Rd	6.1	0.49
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.44
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6
Reinforcement: A_s,prov ≥ A_s,req	6.1	0.39
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.84
Bending: M_Ed ≤ M_Rd	6.1	0.37
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.56
Bar spacing at max. moment: s ≤ min(2h, 250)	9.3.1.1(3)	0.6

Embedded pile wall with anchors

Standard: EN 1997-1 §9, EN 1992-1-1, EN 1998-5

Contiguous or secant bored pile wall, cantilever or with several anchor rows: elastic–plastic subgrade reaction analysis (displacements, earth pressures, moments, anchor forces), passive mobilisation, RC pile design and anchors. Interactive 3D model.

Method and assumptions

- Beam on elastic–plastic Winkler springs: at-rest pressure ± k_h u bounded by active and passive; anchors as springs locked off at P_0.
- Factored effects (DA2: characteristic soil, effects × γ_G) or factored strength (DA1-C2, DA3).

Inputs

Input	Unit	Example	Range	Notes
Excavation depth H	m	7	1 – 25	
Embedment below excavation t	m	5	0.5 – …	

Pile diameter D	m	0.6	0.3 – 1.5	
Pile spacing a	m	0.75	0.3 – 3	Contiguous wall: a ≥ D (gap ≤ ~0.15 m in stiff soil); secant: a < D.
Anchor rows	table			Depth [m], Inclination α [°], Horizontal spacing [m], Lock-off load P_0 [kN], Free length [m], Tendon area A_t [mm²], Design resistance R_a,d [kN] — Leave the table empty for a cantilever wall. Strands Ø15.7: 150 mm² each.
Soil φ'k	°	32	15 – 45	
Soil c'k	kN/m²	5	0 – …	
Soil γ	kN/m³	19	10 – …	
Soil γ_sat	kN/m³	20	10 – …	
Groundwater depth behind the wall	m	20	0 – …	In front the water is lowered to the excavation level.
Surcharge behind the wall (variable)	kN/m²	10	0 – …	
Modulus of subgrade reaction k_h	MN/m³	25	2 – 200	Typical: loose sand 10, medium dense 25, dense 50–100, stiff clay 20–40 MN/m³.
Wall friction δ_a/φ'		0.667	0 – 0.667	
Wall friction δ_p/φ'		0.5	0 – 0.5	
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Include seismic design (EN 1998-5)		no		
Design ground acceleration αgR·γI		0.2	0 – 0.6	
Soil factor S		1.2	1 – 1.8	
Concrete class		C30/37	C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Longitudinal bars		12	6 – 40	
Bar Ø		20	16 / 20 / 25 / 32	

Spiral Ø		10	8 / 10 / 12	
Spiral pitch	mm	150	50 – 400	
Cover to bar centre	mm	85	50 – 150	
Bending stiffness factor (cracking)		0.7	0.3 – 1	E_cm I × factor; 0.5–0.7 for cracked RC piles.

Checks performed (worked example)

Check	Clause	U (example)
DA2 embedment: E_p,mob ≤ E_p,d	EN 1997-1 9.7.4	0.38
DA2 anchor row 1: P_d ≤ R_a,d	EN 1997-1 8.5	0.6
DA2 anchor row 2: P_d ≤ R_a,d	EN 1997-1 8.5	0.73
Bending: M_Ed ≤ M_Rd	6.1	0.17
Shear: V_Ed ≤ max(V_Rd,c, V_Rd,s)	6.2	0.25

Foundation settlement

Standard: EN 1997-1 §6.6, Annex F, Annex H

Immediate and consolidation settlement of a rectangular footing or raft on layered soil: Boussinesq stress distribution, oedometer modulus or C_c/C_r/OCR per layer, influence depth and limits of EN 1997-1 Annex H. 3D stress bulb.

Method and assumptions

- Boussinesq stresses below the centre (Newmark), sub-layers of 0.25 m down to Δσ = 0.2 σ'_v0; granular by E_oed, clays by C_c/C_r/OCR.

Inputs

Input	Unit	Example	Range	Notes
Foundation width B	m	2.5	0.3 – …	
Foundation length L	m	2.5	0.3 – …	
Founding depth D	m	1	0 – …	
Contact pressure (SLS, quasi-permanent)	kN/m²	180	0 – …	Total pressure under the foundation for G + ψ2Q. The net pressure q − γD causes settlement.
Rigid foundation (s = 0.8 s_centre)		yes		A rigid footing settles uniformly ≈ 0.8 × the flexible centre value.
Groundwater depth below ground	m	4	0 – …	

Soil layers (from the ground surface down)	table			Thickness [m], Model, γ [kN/m³], E_oed (granular) [MPa], C_c (clay), C_r (clay), e_0, OCR
Limiting settlement	mm	50	5 – …	EN 1997-1 Annex H: up to 50 mm is often acceptable for normal structures on separate foundations; less for sensitive structures.

Checks performed (worked example)

Check	Clause	U (example)
Settlement: s ≤ s_lim	EN 1997-1 Annex H	0.34

Gabion wall

Standard: EN 1997-1, EN 1998-5, EN 10223-3

Gravity wall of stacked gabion courses with set-backs: Coulomb and Mononobe-Okabe thrust, sliding, overturning, bearing and the stability of every joint between courses. Interactive 3D model.

Method and assumptions

- Global stability as a gravity wall (γ_g = γ_s(1 − n)) plus sliding and eccentricity at every joint between courses.

Inputs

Input	Unit	Example	Range	Notes
Gabion courses (from the bottom up)	table			Width [m], Height [m], Front set-back from the course below [m] — Standard baskets are 0.5 m or 1.0 m high and 1.0–1.5 m wide modules.
Embedment of the base below ground in front	m	0.5	0 – …	
Wall length (3D view)	m	8	1 – …	
Backfill slope β	°	0	0 – 40	
Backfill φ'k	°	30	15 – 45	
Backfill γ	kN/m³	18	10 – …	
Backfill γsat	kN/m³	20	10 – …	
Surcharge on backfill (variable)	kN/m²	10	0 – …	e.g. 10 kN/m² for light traffic or storage.
Water level above base underside	m	0	0 – …	0 = drained backfill (no water pressure).
Foundation φ'k	°	32	10 – 45	

Foundation c'k	kN/m²	0	0 – …	
Foundation γ	kN/m³	19	10 – …	
Base friction factor k (tan δ = k tan φ')		1	0.3 – 1	1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)).
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	'National Annex' uses the approach of the annex chosen in the project details.
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4.
Displacement factor r		2.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped).
ψ2 for surcharge		0.3	0 – 1	
Unit weight of the stone	kN/m³	26	18 – 30	Limestone 24–26, basalt/granite 26–28 kN/m³.
Porosity of the fill n		0.35	0.25 – 0.45	0.30–0.40 for hand-placed, well-graded stone.
Friction angle between courses	°	35	20 – 45	Stone-on-stone contact through the mesh; 30–40°.

Checks performed (worked example)

Check	Clause	U (example)
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.64
DA2 eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.29
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.32
Overturning: M_dst,d ≤ M_stb,d	EN 1997-1 Eq. 2.4	0.31
Seismic sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.8
Seismic eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.33
Seismic bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.59
Seismic overturning: M_dst ≤ M_stb	EN 1998-5 7.3.1	0.29
Joint 1 sliding: H_Ed ≤ N_fav tan φ_j	EN 1997-1 6.5.3	0.42
Joint 1 eccentricity: |e| ≤ b_j/3	EN 1997-1 6.5.4	0.03
Joint 2 sliding: H_Ed ≤ N_fav tan φ_j	EN 1997-1 6.5.3	0.33
Joint 2 eccentricity: |e| ≤ b_j/3	EN 1997-1 6.5.4	0.05

Gravity retaining wall

Standard: EN 1997-1, EN 1998-5, EN 1992-1-1 §12

Mass concrete or masonry gravity wall with optional footing: Coulomb and Mononobe-Okabe thrust, groundwater, sliding, overturning, bearing and plain-concrete section checks. Interactive 3D model.

Method and assumptions

- Same stability engine as the cantilever wall; plain-concrete body sections to EN 1992-1-1 §12.

Inputs

Input	Unit	Example	Range	Notes
Wall height (above footing)	m	3	0.5 – 10	
Top width	m	0.5	0.2 – …	
Width at base of wall body	m	1.9	0.3 – …	
Front face batter (horizontal offset)	m	0.2	0 – …	0 = vertical front face.
Footing thickness (0 = no footing)	m	0.4	0 – …	
Footing toe projection	m	0.5	0 – …	
Footing heel projection	m	0.3	0 – …	
Soil depth over toe	m	0.5	0 – …	
Wall length (3D view)	m	8	1 – …	
Backfill slope β	°	0	0 – 40	
Backfill φ'k	°	30	15 – 45	
Backfill γ	kN/m³	18	10 – …	
Backfill γsat	kN/m³	20	10 – …	
Surcharge on backfill (variable)	kN/m²	10	0 – …	e.g. 10 kN/m² for light traffic or storage.
Water level above base underside	m	0	0 – …	0 = drained backfill (no water pressure).
Foundation φ'k	°	32	10 – 45	
Foundation c'k	kN/m²	0	0 – …	
Foundation γ	kN/m³	19	10 – …	

Base friction factor k (tan δ = k tan φ')		1	0.3 – 1	1.0 cast in situ, 2/3 precast (EN 1997-1 6.5.3(10)).
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	'National Annex' uses the approach of the annex chosen in the project details.
Include seismic design (EN 1998-5)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	EN 1998-1 Table 3.2: ground A 1.0, B 1.2, C 1.15, D 1.35, E 1.4.
Displacement factor r		2.0	1.0 / 1.5 / 2.0	EN 1998-5 Table 7.1: 2.0 free gravity walls allowing ~300α·S mm, 1.0 for walls that cannot move (e.g. propped).
ψ2 for surcharge		0.3	0 – 1	
Concrete class		C20/25	C12/15 / C16/20 / C20/25 / C25/30 / C30/37	
Unit weight of wall	kN/m³	24	15 – …	24 plain concrete, 22 stone masonry, 20 gabions (approx.).

Checks performed (worked example)

Check	Clause	U (example)
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.68
DA2 eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.37
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.42
Overturning: M_dst,d ≤ M_stb,d	EN 1997-1 Eq. 2.4	0.39
Seismic sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.83
Seismic eccentricity: |e| ≤ B/3	EN 1997-1 6.5.4	0.42
Seismic bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.72
Seismic overturning: M_dst ≤ M_stb	EN 1998-5 7.3.1	0.36
Body at base of wall body: N_Ed ≤ N_Rd	12.6.1	0.01
Body at base of wall body: |e| ≤ 0.3 b_w	12.6.5.1	0.64
Body at base of wall body: H_Ed ≤ N_Ed tan 35° (joint)	12.6.3	0.46
Body at mid-height: N_Ed ≤ N_Rd	12.6.1	0.01
Body at mid-height: |e| ≤ 0.3 b_w	12.6.5.1	0.55
Body at mid-height: H_Ed ≤ N_Ed tan 35° (joint)	12.6.3	0.45

Pad footing

Standard: EN 1997-1, EN 1992-1-1, EN 1998-5

Isolated reinforced concrete footing under a column with biaxial moments: effective-area bearing (Annex D, shape and inclination factors), sliding, eccentricity, bending, beam shear and punching. Interactive 3D model.

Method and assumptions

- Effective area B'×L' for biaxial eccentricity (Annex D); punching with the critical perimeter search between 0 and 2d (6.4.4).

Inputs

Input	Unit	Example	Range	Notes
Footing length Bx	m	2.4	0.5 – …	
Footing width By	m	2.4	0.5 – …	
Footing thickness h	m	0.6	0.25 – …	
Founding depth D (ground to underside)	m	1.2	0.3 – …	
Column size cx	m	0.4	0.15 – …	
Column size cy	m	0.4	0.15 – …	
Axial load — permanent G	kN	700	0 – …	Loads at the top of the footing (column base), downward positive.
Axial load — variable Q	kN	300	0 – …	
Moment Mx — permanent	kNm	30		Mx causes eccentricity along X (rotation about the Y axis).
Moment Mx — variable	kNm	20		
Moment My — permanent	kNm	0		My causes eccentricity along Y.
Moment My — variable	kNm	10		
Horizontal load Hx — permanent	kN	10		
Horizontal load Hx — variable	kN	10		
Include seismic combination		yes		
Axial load N_E	kN	750	0 – …	Design values from the seismic analysis (G + ψ2Q + E), capacity

				design as required by EN 1998-1 4.4.2.6.
Moment Mx,E	kNm	150		
Moment My,E	kNm	50		
Horizontal Hx,E	kN	80		
Soil φ'k	°	30	10 – 45	
Soil c'k	kN/m²	5	0 – …	
Soil γ	kN/m³	19	10 – …	
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Concrete class		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover (bottom)	mm	50	25 – …	
Bars in X Ø		14	10 / 12 / 14 / 16 / 20 / 25	
Spacing in X	mm	150	75 – 400	
Bars in Y Ø		14	10 / 12 / 14 / 16 / 20 / 25	
Spacing in Y	mm	150	75 – 400	

Checks performed (worked example)

Check	Clause	U (example)
DA2 eccentricity: $(e_x/B_x)^2 + (e_y/B_y)^2 \le 1/9$	EN 1997-1 6.5.4	0
DA2 bearing: $q_{Ed} \le q_{Rd}$	EN 1997-1 6.5.2	0.36
DA2 sliding: $H_d \le R_{h,d}$	EN 1997-1 6.5.3	0.06
Seismic eccentricity: $(e_x/B_x)^2 + (e_y/B_y)^2 \le 1/9$	EN 1997-1 6.5.4	0.08
Seismic bearing: $q_{Ed} \le q_{Rd}$	EN 1997-1 6.5.2	0.42
Seismic sliding: $H_d \le R_{h,d}$	EN 1997-1 6.5.3	0.19
Reinforcement: $A_{s,prov} \ge A_{s,req}$	6.1	0.6
Minimum reinforcement: $A_{s,prov} \ge A_{s,min}$	9.2.1.1	0.71
Bending: $M_{Ed} \le M_{Rd}$	6.1	0.58
Shear: $V_{Ed} \le V_{Rd,c}$	6.2.2	0.64
Reinforcement: $A_{s,prov} \ge A_{s,req}$	6.1	0.55
Minimum reinforcement: $A_{s,prov} \ge A_{s,min}$	9.2.1.1	0.69

Bending: M_Ed ≤ M_Rd	6.1	0.54
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.61
Punching: v_Ed ≤ v_Rd	6.4.4(2)	0.58
Punching at column face: v_Ed,0 ≤ v_Rd,max	6.4.5(3)	0.41

Single pile — axial and lateral

Standard: EN 1997-1 §7, EN 1992-1-1

Bored, CFA or driven pile in layered clay/sand: shaft and base resistance from soil parameters (α-method, β-method, Meyerhof N_q*), partial and correlation/model factors of the National Annex, tension, lateral resistance (Broms) and structural check. Interactive 3D model.

Method and assumptions

- Shaft: α-method (clay) and β-method K tan δ σ'_v (sand); base: 9c_u or Meyerhof N_q* with base efficiency η_b (0.5 bored/CFA, 1.0 driven).
- Characteristic value by model factor (7.6.2.3(8)) or ξ3/ξ4; lateral by Broms (free head, top layer).

Inputs

Input	Unit	Example	Range	Notes
Pile type		Bored	Bored / CFA / Driven	
Pile diameter D	m	0.8	0.2 – 3	
Pile length L (below ground)	m	18	2 – 80	
Concrete class		C30/37	C20/25 / C25/30 / C30/37 / C35/45 / C40/50	
Reinforcement fyk	N/mm²	500	400 – 600	
Longitudinal bars		10	6 – 40	
Bar Ø		20	12 / 14 / 16 / 20 / 25	
Cover to bar centre	mm	75	40 – 150	Bored piles: cover 75 mm to the links is usual (EN 1536).
Axial load — permanent G	kN	800	0 – …	

Axial load — variable Q	kN	250	0 – …	
Uplift (tension) load — variable	kN	0	0 – …	e.g. wind uplift; checked against shaft resistance in tension.
Horizontal load — permanent	kN	20	0 – …	
Horizontal load — variable	kN	30	0 – …	
Height of the horizontal load above ground e	m	0	0 – …	
Groundwater depth below ground	m	3	0 – …	
Soil layers (from the ground surface down)	table			Thickness [m], Soil, γ [kN/m³], φ' (sand) [°], c_u (clay) [kN/m²]
Adhesion factor α (clay)		0.5	0.2 – 1	Typical 0.45–0.5 for bored piles in stiff clay; 0.5–1.0 driven.
Characteristic resistance		Alternative method (model factor)	Alternative method (model factor) / Ground test profiles (ξ3, ξ4)	Alternative method: R_k = R_cal / γ_Rd (7.6.2.3(8)). Ground test profiles: R_k = min(R_mean/ξ3, R_min/ξ4) (7.6.2.3(5)).
Model factor γ_Rd		1.4	1 – 2	Cyprus NA 2.19: at least 1.25. About 1.4 when no load tests are made.
Number of test profiles n		1	1 – 10	
Stiff pile cap (ξ / 1.1, 7.6.2.3(7))		no		
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	

Checks performed (worked example)

Check	Clause	U (example)
DA2 compression: F_c,d ≤ R_c,d	7.6.2.1	0.8
DA2 lateral: H_d ≤ H_Rd	EN 1997-1 7.7	0.29
Minimum reinforcement A_s ≥ 0.5 % A_c (A_c ≤ 0.5 m²)	EN 1992-1-1 9.8.5 Table 9.6N	0.76

Axial: N_Ed ≤ N_Rd	6.1	0.14
Bending with axial force: M_Ed ≤ M_Rd(N_Ed)	6.1	0.17

Slope stability (Bishop)

Standard: EN 1997-1 §11, EN 1998-5 §4.1.3

Circular slip surfaces by Bishop's simplified method with automatic search of the critical circle; pore pressure ratio, crest surcharge and seismic pseudo-static action. Interactive 3D model.

Method and assumptions

- Bishop simplified method with automatic search of the critical circle; pseudo-static seismic action (EN 1998-5 4.1.3.3).

Inputs

Input	Unit	Example	Range	Notes
Slope height	m	6	1 – 50	
Slope angle	°	30	5 – 75	Angle of the slope face to the horizontal (1:√3 ≈ 30°, 1:2 ≈ 26.6°).
Depth to firm stratum below toe	m	6	0 – …	Slip circles are not allowed below this level.
Slope length (3D view)	m	20	4 – …	
Soil φ'k	°	30	0 – 45	
Soil c'k	kN/m²	8	0 – …	
Soil γ	kN/m³	19	10 – …	
Pore pressure ratio r_u		0	0 – 0.6	u = r_u·γ·h. 0 = dry/drained, 0.2–0.3 typical with a high water table.
Crest surcharge (variable)	kN/m²	10	0 – …	
Distance of surcharge from crest	m	1	0 – …	
Surcharge width	m	5	0 – …	
Include seismic (pseudo-static)		yes		
Design ground acceleration αgR·γI		0.16	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Soil factor S		1.2	1 – 1.8	
Topography factor S_T		1	1 – 1.4	EN 1998-5 Annex A: 1.0 for slopes < 15°, up to 1.2–1.4 for high ridges.

Checks performed (worked example)

Check	Clause	U (example)
Persistent: E_d ≤ R_d (1 ≤ F_d)	EN 1997-1 Eq. 2.5	0.8
Seismic: E_d ≤ R_d (1 ≤ F_d,E)	EN 1998-5 4.1.3.3	0.81

Strip footing and tie beams

Standard: EN 1997-1, EN 1992-1-1, EN 1998-5

Continuous footing under a wall (per metre run): bearing with load inclination, sliding, eccentricity, transverse bending and shear. Optional seismic tie beams between footings (EN 1998-5 5.4.1.2).

Method and assumptions

- Strip bearing per metre run with load inclination; tie-beam force ±k α S N (EN 1998-5 5.4.1.2) with the minimum sections of EN 1998-1 5.8.2.

Inputs

Input	Unit	Example	Range	Notes
Footing width B	m	1.4	0.4 – …	
Footing thickness h	m	0.45	0.2 – …	
Founding depth D (ground to underside)	m	1	0.3 – …	
Wall thickness b_w	m	0.25	0.1 – …	
Line load — permanent G	kN/m	180	0 – …	At the top of the footing (base of the wall), downward positive.
Line load — variable Q	kN/m	60	0 – …	
Moment — permanent	kNm/m	5		Moment about the footing axis (across the width B).
Moment — variable	kNm/m	5		
Horizontal load — permanent	kN/m	0		
Horizontal load — variable	kN/m	5		
Include seismic combination		no		
Line load N_E	kN/m	200	0 – …	G + ψ2Q + E from the seismic analysis.
Moment M_E	kNm/m	30		
Horizontal H_E	kN/m	25		

Soil φ'k	°	30	10 – 45	
Soil c'k	kN/m²	5	0 – …	
Soil γ	kN/m³	19	10 – …	
Design approach		National Annex	National Annex / DA1 / DA2 / DA3	
Concrete class		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Reinforcement fyk	N/mm²	500	400 – 600	
Cover (bottom)	mm	50	25 – …	
Transverse bars Ø		12	10 / 12 / 14 / 16 / 20 / 25	
Spacing of transverse bars	mm	200	75 – 400	
Design tie beams (seismic)		yes		Tie beams connect the footings so they move together in an earthquake.
Reference ground acceleration a_gR		0.25	0 – 0.6	In g. Use the zone buttons.
Ground type		C	B / C / D	Ground type A: tie beams may be omitted (EN 1998-5 5.4.1.2(5)).
Importance class		II	I / II / III / IV	
Design axial load of column/wall 1	kN	900	0 – …	Seismic design situation, the two elements the beam joins.
Design axial load of column/wall 2	kN	700	0 – …	
Tie beam width	m	0.3	0.15 – …	
Tie beam depth	m	0.5	0.2 – …	
Storeys above the basement		3	1 – 40	
Bars top + bottom (total)		8	4 – 16	Half at the top, half at the bottom.
Tie beam bars Ø		16	10 / 12 / 14 / 16 / 20 / 25	

Checks performed (worked example)

Check	Clause	U (example)
DA2 eccentricity: e/B ≤ 1/3	EN 1997-1 6.5.4	0.1
DA2 bearing: q_Ed ≤ q_Rd	EN 1997-1 6.5.2	0.56
DA2 sliding: H_d ≤ R_h,d	EN 1997-1 6.5.3	0.07

Reinforcement: A_s,prov ≥ A_s,req	6.1	0.5
Minimum reinforcement: A_s,prov ≥ A_s,min	9.2.1.1	0.93
Bending: M_Ed ≤ M_Rd	6.1	0.48
Shear: V_Ed ≤ V_Rd,c	6.2.2	0.33
Tie beam width b ≥ 0.25 m	EN 1998-1 5.8.2(4)	0.83
Tie beam depth h ≥ 0.4 m (3 storeys)	EN 1998-1 5.8.2(4)	0.8
Minimum ratio top and bottom ρ_b ≥ 0.4 %	EN 1998-1 5.8.2(5)	0.75
Tie beam in tension: N_tie ≤ N_t,Rd	EN 1992-1-1 6.1	0.13
Tie beam in compression: N_tie ≤ N_c,Rd	EN 1992-1-1 6.1	0.03

LOADS

Load combinations

Standard: EN 1990 Annex A1

Generates all ULS (6.10), seismic (6.12b) and SLS (characteristic, frequent, quasi-permanent) combinations from your actions, with the ψ factors of the National Annex, and finds the governing values.

Method and assumptions

- All ULS (6.10), seismic (6.12b) and SLS combinations with ψ of the annex; each variable action in turn as leading action.

Inputs

Input	Unit	Example	Range	Notes
Actions (characteristic values, one effect e.g. N, M or V)	table			Name, Type, Category (imposed), Value
Site altitude (for snow ψ)		100	0 – …	m
Unit of the values		kN	kN / kNm / kN/m / kN/m²	

Seismic spectrum and lateral forces

Standard: EN 1998-1

Design response spectrum (Type 1) and the lateral force method: fundamental period, base shear and storey forces for regular buildings.

Method and assumptions

- Type 1 design spectrum, T1 = C_t H^¾, λ = 0.85 when applicable, storey forces with a linear mode shape (Eq. 4.11).

Inputs

Input	Unit	Example	Range	Notes
Reference ground acceleration a_gR		0.25	0 – 0.6	In g (a_gR × γI). Pick a zone with the buttons of the selected National Annex.
Ground type		B	A / B / C / D / E	A rock, B very dense sand/gravel or stiff clay, C dense to medium-dense, D loose, E soft surface layer on rock.
Importance class		II	I / II / III / IV	II ordinary buildings, III schools/assembly, IV hospitals/civil protection.
Behaviour factor q		3.9	1 – 8	e.g. RC frame DCM multi-storey: 3.0·αu/α1 = 3.9.
Structural system (for T1 = C_t H^¾)		RC moment frames / eccentric braced steel	Steel moment frames / RC moment frames / eccentric braced steel / All other structures (walls, masonry…)	
Fundamental period T1 (0 = estimate)		0	0 – 4	s
Storeys (height and weight)	table			Storey height [m], Seismic weight G + ψE Q [kN]

Checks performed (worked example)

Check	Clause	U (example)
Lateral force method: T_1 ≤ min(4T_C, 2.0 s)	4.3.3.2.1(2)	0.24

Wind and snow loads at the site

Standard: EN 1991-1-4, EN 1991-1-3

Peak velocity pressure q_p(z), wall pressures and roof snow load. With the Cyprus National Annex v_b,0 and s_k are taken automatically from the project site on the official maps.

Method and assumptions

- q_p(z) by Eq. 4.8 with c_dir = c_season = 1; walls by Table 7.1; roof snow μ_1 C_e C_t s_k.

Inputs

Input	Unit	Example	Range	Notes
Use the project site (maps of the National Annex)		yes		Pick the site in the project details (📍 Map). Otherwise give v_b,0 and s_k.
Fundamental basic wind velocity v_b,0		24	10 – 60	m/s — used when no site is selected or for EN values.
Site altitude above sea level		100	0 – 3000	m
Ground snow load s_k (manual)	kN/m²	0.5	0 – …	Used only when the National Annex gives no snow map/formula.
Terrain category		III	0 / I / II / III / IV	0 sea, I lakes/flat, II open country, III suburbs/villages, IV cities (≥15 % covered by buildings > 15 m).
Orography factor c_o		1	1 – 2	1.0 unless the site is on a hill or cliff (Annex A.3).
Reference height z (building height)	m	9	1 – 200	
Building depth in the wind direction d	m	12	1 – …	
Roof pitch α	°	0	0 – 80	
Exposure (snow)		Normal	Windswept / Normal / Sheltered	

MASONRY

Masonry wall

Standard: EN 1996-1-1

Load-bearing unreinforced masonry wall: f_k from units and mortar, γ_M by category and class, effective height and thickness, slenderness, reduction factors at top/bottom/mid-height (Annex G) and in-plane shear of a shear wall. Interactive 3D model.

Method and assumptions

- Vertical resistance with the reduction factors Φ at top, bottom and mid-height (Annex G, E = K_E f_k); in-plane shear by Eq. 6.13.

Inputs

Input	Unit	Example	Range	Notes
Masonry unit		Clay	Clay / Calcium silicate / Aggregate concrete / Autoclaved aerated concrete / Manufactured stone / Dimensioned natural stone	
Unit group		Group 2	Group 1 / Group 2 / Group 3 / Group 4	Group 1 solid (voids ≤ 25 %), 2 vertically perforated ≤ 55 %, 3 ≤ 70 %, 4 horizontally perforated.
Normalised unit strength f_b	N/mm²	10	2 – 60	
Mortar class		M10	M2.5 / M5 / M10 / M15 / M20	
Units and mortar		Category I units, designed mortar	Category I units, designed mortar / Category I units, prescribed mortar / Category II units, any mortar	
Execution control class		2	1 / 2 / 3 / 4 / 5	1 = best supervision … 5 = least.
Final creep coefficient φ∞		1	0.3 – 3	Clay 0.5–1.5, concrete/calcium silicate 1.0–2.0 (Table 6 CYS).
Wall thickness t	mm	250	90 – 600	
Clear storey height h	m	3	1 – 8	
Wall length l_w	m	4	0.3 – …	

Restraint of the wall		Top and bottom (RC floors)	Top and bottom (RC floors) / Top and bottom (timber floors) / Top, bottom and one vertical edge / Top, bottom and both vertical edges	
Design vertical load at the top N_Ed	kN/m	180	0 – …	
Eccentricity of the floor load at the top	mm	15	0 – …	From the floor bearing (e.g. t/6 for a slab bearing on part of the wall).
In-plane shear on the wall V_Ed	kN	60	0 – …	0 = not a shear wall.
Unit weight of masonry	kN/m³	18	5 – 25	

Checks performed (worked example)

Check	Clause	U (example)
Slenderness: h_ef/t_ef ≤ 27	5.5.1.4	0.33
Vertical load at top: N_Ed ≤ Φ t f_d	Eq. 6.2	0.32
Vertical load at mid-height: N_Ed ≤ Φ t f_d	Eq. 6.2	0.34
Vertical load at bottom: N_Ed ≤ Φ t f_d	Eq. 6.2	0.33
In-plane shear: V_Ed ≤ V_Rd	6.2	0.17

STEEL

Bolted connection

Standard: EN 1993-1-8 §3

Bolt group in single or double shear with eccentric shear and tension: spacing rules, bolt shear, bearing, tension, punching, combined shear and tension, long joints, block tearing and the net section of the plate. Interactive 3D model with bolt forces.

Method and assumptions

- Elastic distribution of eccentric shear in the bolt group; bearing checked for the governing bolt with the end/inner values of Table 3.4.

Inputs

Input	Unit	Example	Range	Notes
Bolt class		8.8	4.6 / 5.6 / 8.8 / 10.9	
Bolt diameter		M20	8 …	
Threads in the shear plane		yes		
Shear planes		1	1 / 2	2 = double cover plates (splice).
Bolt rows in the force direction n1		3	1 – 12	
Bolt columns across n2		2	1 – 8	
Pitch p1 (in force direction)	mm	70	20 – …	
Pitch p2 (across)	mm	70	20 – …	
End distance e1	mm	50	10 – …	
Edge distance e2	mm	40	10 – …	
Thinnest plate thickness t	mm	12	4 – 60	
Plate steel grade		S275	S235 / S275 / S355	
Shear force V_Ed (along the force direction)	kN	300	0 – …	
Eccentricity of V from the bolt-group centroid	mm	0	0 – …	Causes a torsional moment on the group (elastic method).
Tension per bolt F_t,Ed	kN	0	0 – …	

Checks performed (worked example)

Check	Clause	U (example)
e_1 ≥ 1.2 d_0	Table 3.3	0.53
e_2 ≥ 1.2 d_0	Table 3.3	0.66
p_1 ≥ 2.2 d_0	Table 3.3	0.69
p_2 ≥ 2.4 d_0	Table 3.3	0.75
p_1, p_2 ≤ min(14t, 200)	Table 3.3	0.42
Bolt shear: F_v,Ed ≤ F_v,Rd	Table 3.4	0.53
Bearing: F_v,Ed ≤ F_b,Rd	Table 3.4	0.32
Group: V_Ed ≤ n min(F_v,Rd, F_b,Rd)	3.7	0.53
Plate tension: V_Ed ≤ min(N_pl,Rd, N_u,Rd)	EN 1993-1-1 6.2.3	0.76
Block tearing: V_Ed ≤ V_eff,1,Rd	3.10.2	0.42

Column base plate

Standard: EN 1993-1-8 §6.2.5–6.2.8, EN 1992-1-1 §6.7

Base plate of an I/H column on a concrete foundation under axial force, strong-axis moment and shear: bearing strength f_jd, effective width c, compression T-stubs, tension T-stubs with anchor bolts, moment resistance and shear by friction and bolts. Interactive 3D model.

Method and assumptions

- Component method: compression T-stubs on the foundation, tension T-stub with anchor bolts (no prying), M–N from statics about the compressed flange.
- Anchorage of the bolts in concrete (EN 1992-4) and welds are not checked.

Inputs

Input	Unit	Example	Range	Notes
Column section		HEB 240	40 …	
Axial compression N_Ed	kN	600	0 – …	
Moment M_y,Ed	kNm	100	0 – …	
Shear V_Ed	kN	40	0 – …	
Plate width b_p (along the flanges)	mm	400	150 – …	
Plate length l_p (along the web)	mm	500	150 – …	
Plate thickness t_p	mm	25	8 – 80	
Plate steel grade		S275	S235 / S275 / S355	
Anchor bolt class		8.8	4.6 / 5.6 / 8.8 / 10.9	
Anchor bolt diameter		M24	7 …	
Bolts per side (in each tension row)		2	1 – 4	
Bolt distance from plate end e_x	mm	50	25 – …	
Bolt gauge across w	mm	240	60 – …	
Foundation concrete		C25/30	C20/25 / C25/30 / C30/37 / C35/45	
Foundation width	m	1.2	0.3 – …	
Foundation length	m	1.2	0.3 – …	
Flange weld throat a	mm	8	3 – 20	

Checks performed (worked example)

Check	Clause	U (example)
Pure compression: N_Ed ≤ N_j,Rd	6.2.8.2	0.28
Tension side: F_t,Ed ≤ F_T,Rd	Table 6.7	0.44
Compression side: F_c,Ed ≤ F_C,Rd	Table 6.7	0.69

Shear: V_Ed ≤ F_f,Rd + F_vb,Rd	6.2.2(8)	0.12

Steel beam design

Standard: EN 1993-1-1:2005

Simply supported rolled I-beam: classification, bending, shear, bending-shear interaction, lateral-torsional buckling and deflection.

Method and assumptions

- Simply supported rolled I-beam; LTB with M_cr (NCCI SN003) and 6.3.2.3.

Inputs

Input	Unit	Example	Range	Notes
Section		IPE 300	40 …	
Steel grade		S275	S235 / S275 / S355 / S460	
Span	m	6	0.1 – …	
LTB effective length (0 = unrestrained span)	m	2	0 – …	Distance between lateral restraints to the compression flange.
Permanent UDL (excl. self weight)	kN/m	8	0 – …	
Variable UDL	kN/m	10	0 – …	
Permanent point load at midspan	kN	0	0 – …	
Variable point load at midspan	kN	0	0 – …	
γG		1.35	0 – …	
γQ		1.5	0 – …	
γM0		1	0.5 – …	
γM1		1	0.5 – …	
Variable load deflection limit L/…		360	1 – …	

Checks performed (worked example)

Check	Clause	U (example)
Bending: M_Ed ≤ M_c,Rd	6.2.5(1)	0.69
Shear: V_Ed ≤ V_pl,Rd	6.2.6(1)	0.19
Shear buckling: h_w/t_w ≤ 72ε/η	6.2.6(6)	0.59

LTB: M_Ed ≤ M_b,Rd	6.3.2.1(1)	0.73
Deflection: δ_q ≤ δ_lim	EN 1990 A1.4	0.58

Steel column

Standard: EN 1993-1-1 §6.3, Annex B

Rolled I/H column under axial force and biaxial end moments: classification, cross-section resistance, flexural buckling about both axes, lateral-torsional buckling and the interaction of Eq. 6.61/6.62 (Annex B). Interactive 3D model with buckled shape.

Method and assumptions

- Buckling curves of Table 6.2; interaction factors of Annex B (Method 2).
- Class 4 webs by the effective width of EN 1993-1-5.

Inputs

Input	Unit	Example	Range	Notes
Section		HEB 240	40 …	
Steel grade		S355	S235 / S275 / S355 / S460	
Column height L	m	4	0.5 – 30	
Buckling length factor about y-y		1	0.5 – 3	1.0 pinned–pinned, 0.7 fixed–pinned, 0.5 fixed–fixed, 2.0 cantilever; sway frames > 1.0.
Buckling length factor about z-z		1	0.3 – 3	Bracing at mid-height about the weak axis → 0.5.
LTB length factor		1	0.3 – 3	
Axial compression N_Ed	kN	1200	0 – …	
M_y,Ed at top	kNm	80		Strong-axis moment; same sign at both ends = single curvature.
M_y,Ed at bottom	kNm	-40		
M_z,Ed at top	kNm	5		
M_z,Ed at bottom	kNm	0		
Shear V_z,Ed	kN	30	0 – …	

Checks performed (worked example)

Check	Clause	U (example)
Cross-section: N/N_Rd + M_y/M_y,Rd + M_z/M_z,Rd ≤ 1	Eq. 6.2	0.56
Shear: V_Ed ≤ V_pl,Rd	6.2.6	0.04

Buckling about y-y: N_Ed ≤ N_b,Rd	6.3.1	0.36
Buckling about z-z: N_Ed ≤ N_b,Rd	6.3.1	0.51
LTB: M_y,Ed ≤ M_b,Rd	6.3.2.1	0.21
Eq. 6.61: N/(χ_y N_Rk/γ_M1) + k_yy M_y/(χ_LT M_y,Rk/γ_M1) + k_yz M_z/M_z,Rd ≤ 1	6.3.3(4)	0.47
Eq. 6.62: N/(χ_z N_Rk/γ_M1) + k_zy M_y/(χ_LT M_y,Rk/γ_M1) + k_zz M_z/M_z,Rd ≤ 1	6.3.3(4)	0.69

TIMBER

Timber beam

Standard: EN 1995-1-1, EN 338, EN 14080

Simply supported solid timber or glulam beam: k_mod by service class and load duration, size factor, bending with lateral-torsional buckling (k_crit), shear (k_cr), bearing (k_c,90) and instantaneous/final deflection with creep (k_def). Interactive 3D model.

Method and assumptions

- Governing combination by the largest w_d / k_mod; deflection includes shear deformation and creep (k_def).

Inputs

Input	Unit	Example	Range	Notes
Strength class		GL24h	8 …	
Width b	mm	140	38 – 400	
Depth h	mm	400	75 – 2000	
Span L	m	6	0.5 – 40	
Lateral restraint spacing (0 = span)	m	0	0 – …	Distance between restraints of the compression edge (e.g. purlins, floor joists).
Load on the compression (top) edge		yes		Destabilising: l_ef + 2h (Table 6.1).
Support length	mm	120	40 – …	
Service class		1	1 / 2 / 3	1 heated interior, 2 covered unheated, 3 exposed outdoors.
Permanent load g_k (excl. self weight)	kN/m	2	0 – …	
Variable load q_k	kN/m	3	0 – …	
Duration of the variable load		Medium-term	Permanent / Long-term / Medium-term / Short-term /	Imposed floor load: medium; snow: short (Cyprus); wind:

			Instantaneous	short/instant.
ψ2 of the variable load		0.3	0 – 1	
Limit w_inst (variable) = L / …		300	100 – …	
Limit w_net,fin = L / …		250	100 – …	

Checks performed (worked example)

Check	Clause	U (example)
Bending: σ_m,d ≤ f_m,d	Eq. 6.11	0.57
LTB: σ_m,d ≤ k_crit f_m,d	Eq. 6.33	0.57
Shear: τ_d ≤ f_v,d	Eq. 6.13	0.4
Bearing: σ_c,90,d ≤ k_c,90 f_c,90,d	Eq. 6.3	0.38
w_inst,Q ≤ L/300	Table 7.2	0.32
w_net,fin ≤ L/250	Table 7.2	0.63